4.4 Electric and Magnetic Fields

This topic covers Coulomb's law, capacitors, magnetic flux density and the laws of electromagnetic induction.

This topic may be studied using applications that relate to, for example, communications and display techniques.

This unit includes many opportunities for developing experimental skills and techniques by carrying out more than just the core practical experiment.

Candidates will be assessed on their ability to:

92	understand that an electric field (force field) is defined as a region where a charged particle experiences a force
93	understand that electric field strength is defined as $E = \frac{F}{Q}$ and be able to use this equation
94	be able to use the equation $F = \frac{Q_1 Q_2}{4\pi\varepsilon_0 r^2}$ for the force between two charges
95	be able to use the equation $E = \frac{Q}{4\pi\varepsilon_0 r^2}$ for the electric field due to a point charge
96	know and understand the relation between electric field and electric potential
97	be able to use the equation $E = \frac{V}{d}$ for an electric field between parallel plates
98	be able to use $V = \frac{Q}{4\pi\varepsilon_0 r}$ for a radial field
99	be able to draw and interpret diagrams using field lines and equipotentials to describe radial and uniform electric fields
100	understand that capacitance is defined as $C = \frac{Q}{V}$ and be able to use this equation
101	be able to use the equation $W = \frac{1}{2}QV$ for the energy stored by a capacitor, be able to derive the equation from the area under a graph of potential difference against charge stored and be able to derive and use the equations $W = \frac{1}{2}CV^2$ and $W = \frac{\frac{1}{2}Q^2}{C}$

102	be able to draw and interpret charge and discharge curves for resistor capacitor circuits and understand the significance of the time constant *RC*
103	**CORE PRACTICAL 11: Use an oscilloscope or data logger to display and analyse the potential difference (p.d.) across a capacitor as it charges and discharges through a resistor**
104	be able to use the equation $Q = Q_0 e^{-t/RC}$ and derive and use related equations for exponential discharge in a resistor-capacitor circuit, $I = I_0\, e^{-t/RC}$, and $V = V_0\, e^{-t/RC}$ and the corresponding log equations $\ln Q = \ln Q_0 - \frac{t}{RC}$, $\ln I = \ln I_0 - \frac{t}{RC}$ and $\ln V = \ln V_0 - \frac{t}{RC}$
105	understand and use the terms *magnetic flux density B, flux* ϕ and *flux linkage N*ϕ
106	be able to use the equation $F = Bqv \sin\theta$ and apply Fleming's left-hand rule to charged particles moving in a magnetic field
107	be able to use the equation $F = BIl \sin\theta$ and apply Fleming's left-hand rule to current carrying conductors in a magnetic field
108	understand the factors affecting the e.m.f. induced in a coil when there is relative motion between the coil and a permanent magnet
109	understand the factors affecting the e.m.f. induced in a coil when there is a change of current in another coil linked with this coil
110	understand how to use Faraday's law to determine the magnitude of an induced e.m.f. and be able to use the equation that combines Faraday's and Lenz's laws $\mathscr{E} = \frac{-\mathrm{d}(N\phi)}{\mathrm{d}t}$.